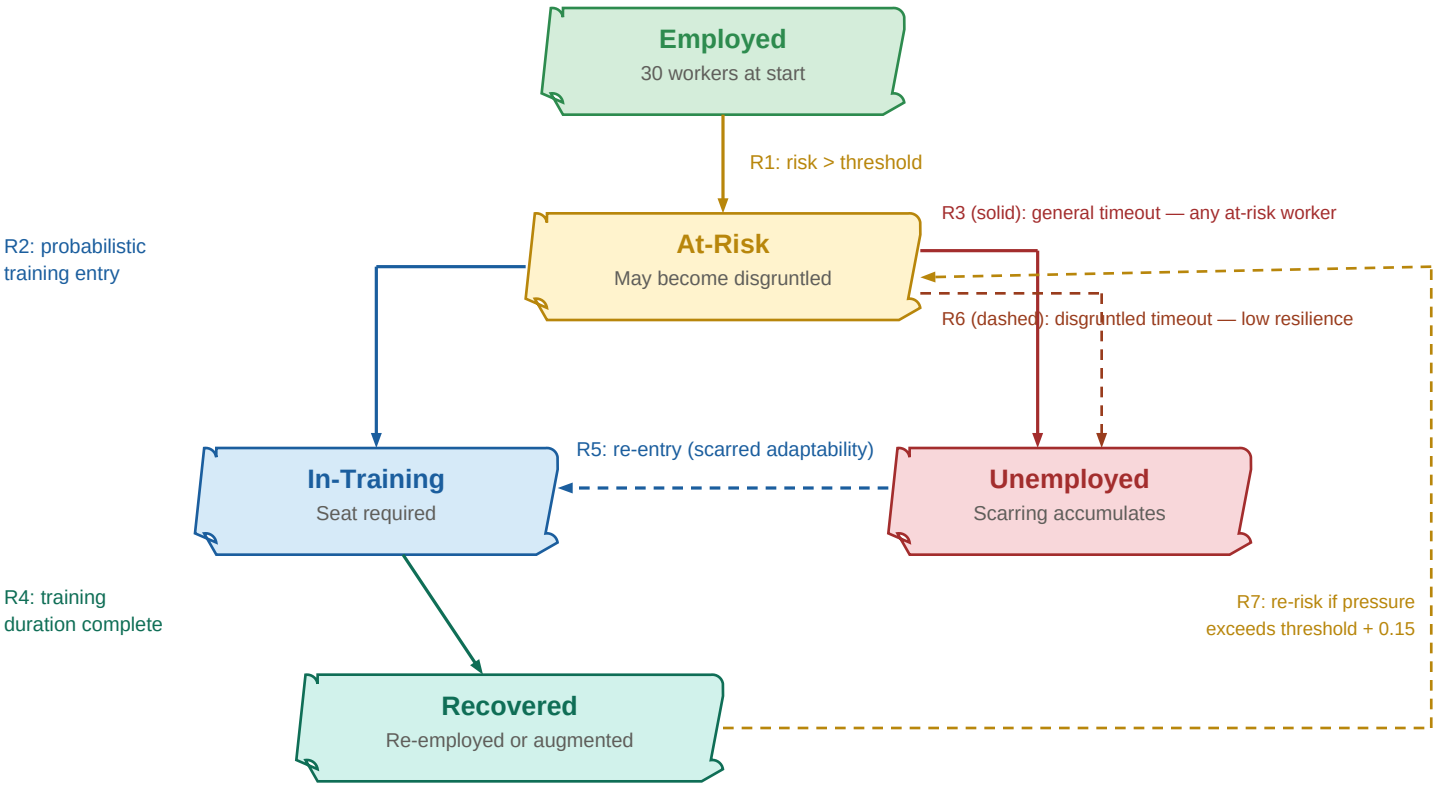
AI Workforce Odyssey — State Transition Diagram

Agent-based model: 7 programmed transition rules + emergent feedback mechanisms



R3 vs R6 — key distinction

R3 (general timeout): Any at-risk worker who doesn't enter training within the at-risk-timeout (6 or 10 ticks) becomes unemployed.

R6 (disgruntled path): Workers with resilience < 40 become disgruntled after 3 ticks (–20 adaptability). After the disgruntled timeout (8 or 12 ticks), they become unemployed. They also erode nearby workers' resilience (–2/tick), creating contagion.

Exit priority: R2 (training) | R3 (general timeout) | R6 (disgruntled timeout) — whichever fires first

Emergent mechanisms (not directly programmed as outcomes)

Peer influence
+5 per nearby training/recovered (cap 20)

Scarring feedback loop
Longer unemp. = lower adaptability

Training bottleneck
Capped seats amplify gaps

Disgruntled contagion
Nearby erode resilience (–2/tick)

Adaptation inequality
Equal access ≠ equal outcomes

Spatial clusters
Recovery vs stagnation zones

Legend

Solid arrow = primary transition (always possible)

Dashed arrow = feedback / conditional transition

States: Employed · At-Risk · In-Training · Unemployed · Recovered

Scenario parameters:

Parameter	Tech-Driven	Human-Centric
Base pressure / Acceleration	10 / 0.40 per tick	5 / 0.20 per tick
Disruption threshold	0.50	0.55
Training seats / Duration	8 seats / 8 ticks	20 seats / 5 ticks
Augmentation probability	30%	55%
At-risk / Disgruntled timeout	6 / 8 ticks	10 / 12 ticks
Peer influence radius	2 patches	4 patches
Scarring rate	0.5 / tick	0.3 / tick

Note: Parameter values are not calibrated to empirical data. They were set through iterative test runs to produce balanced dynamics in both scenarios (see Design Choices doc, #19 and #23). Design concepts—accelerating pressure and junior vulnerability—are informed by Stanford Digital Economy Lab research on AI employment effects.